Krishi Attri

krishiattriwork@gmail.com · linkedin.com/in/krishi-attri15 · github.com/Archerkattri · archerkattri.github.io

Date of Birth: 15 May 2002 · Nationality: Indian

Research Profile

Robotics researcher admitted to the Ph.D. programme at the University of Central Florida (Rehabilitation Engineering & Assistive Device Lab), currently completing an M.S. in Mechanical Engineering at Seoul National University under a Government Science Fellowship (GSFS). My research focuses on **online 3D reconstruction and pose tracking for robotic manipulation**, specifically developing visuo-tactile SLAM systems built around explicit **3D Gaussian Splatting** representations that enable robots to reconstruct and track objects under heavy occlusion during in-hand manipulation.

I am currently developing and integrating these systems as part of a government-funded humanoid robot project at the lab, fusing multimodal perception—RGB-D vision, DIGIT tactile sensing, and hand proprioception—into a shared Gaussian object state that simultaneously supports reconstruction, pose tracking, and manipulation interfaces. My work spans GPU-accelerated algorithm development, multimodal sensor fusion, and full-stack hardware–software integration on platforms including the UR5e and Allegro Hand.

Education

Ph.D., Mechanical Engineering — Post Master's Track *Aug 2026 (Incoming)*
University of Central Florida | ORCGS Doctoral Fellow | Rehabilitation Engineering & Assistive Device Lab (REAL) | Advisor: Professor Hwan Choi *Orlando, FL, USA*

Master of Science in Mechanical Engineering *Sept 2024 – July 2026 (Expected)*
Seoul National University | GSFS Scholar | Soft Robotics & Bionics Laboratory *Seoul, South Korea*

Bachelor of Science in Mechanical Engineering *Aug 2020 – May 2024*
Villanova University | Minor in Mechatronics | Control & Dynamics *Villanova, PA, USA*

Exchange Student — B.S. Mechanical Engineering *Aug 2022 – June 2023*
Yonsei University *Seoul, South Korea*

Research Experience

Graduate Research Student *Sept 2024 – July 2026*
Soft Robotics & Bionics Laboratory, Seoul National University *Seoul, South Korea*
Government-Funded Humanoid Robot Project at the Lab

- Contributing to a multi-year government-funded humanoid robot programme at the Soft Robotics & Bionics Lab (Phase 1 completed 2024; Phase 2 ongoing).
- Integrating visuo-tactile SLAM and dexterous in-hand manipulation modules into the Phase 2 full-scale humanoid prototype.
- Coordinating cross-disciplinary engineering teams to meet hardware–software integration milestones.

GaussianFeels: Real-Time Multimodal 3D Reconstruction with Tactile-Enhanced Gaussian Splatting (M.S. Thesis)
Advisor: Professor Yong-Lae Park, Seoul National University

- Developed GaussianFeels: an online visuo-tactile reconstruction and pose-tracking system built around an explicit object-centric 3D Gaussian Splatting map for robotic in-hand manipulation under heavy occlusion.
- Designed the system around four integrated components: (1) multimodal sensor ingestion synchronising RGB-D frames, segmentation masks, tactile contact geometry (DIGIT sensor), and hand proprioception into a per-frame representation; (2) an object-centric Gaussian map maintained online with contact-aware population management and occlusion-aware multimodal supervision; (3) a pose tracker embedded inside the SLAM loop supporting GT, GT-init, and full SLAM modes; and (4) a parallel manipulation branch for frame-0 shape completion and progressive geometry replacement.
- Implemented contact-aware Gaussian population management: multimodal Gaussian spawning, direct contact-point insertion, tactile-region density boosting, and selective freezing to preserve contact-critical regions under partial visibility.
- Designed an occlusion-aware training objective that dynamically reweights RGB, depth, and tactile supervision based on hand-induced occlusion, maintaining reconstruction fidelity when visual evidence becomes unreliable.
- Implemented a frame-0 shape-completion bootstrap using Hunyuan3D-2-mini: aligns an orientation-variant 3D prior to the real frame-0 observation, then progressively replaces generated geometry with measured camera and tactile data as the episode unfolds.
- Evaluated on the FeelSight family of benchmarks across three dataset variants: simulation, real-world robot rollouts, and occlusion-focused episodes.

PoP-SLAM: Point Cloud Projection for SLAM
Co-authors: Seongmin Jung (SNU AI), Josselin Marchand (École des Mines), Michele Lorenzo Paolicchi (Vrije Universiteit Amsterdam)

- Co-developed PoP-SLAM: a dense visual SLAM system that introduces a projection-first rendering strategy, eliminating computationally intensive nearest-neighbor searches inherent to neural point cloud-based SLAM.
- Designed GPU-vectorised point cloud projection: transforms up to ∼15 000 neural points per frame into image space using vectorised matrix multiplication, enabling real-time SLAM at ∼**4 FPS**.
- Introduced direct occlusion detection via multi-keyframe depth masking: projects neural points to nearby keyframes and retains only points consistent with measured depth across all keyframes, avoiding volume rendering entirely.
- Achieved **best ATE RMSE of 0.75 cm** on TUM-RGBD dataset, outperforming Point-SLAM, NICE-SLAM, ESLAM, and SplaTAM; competitive 0.38 cm average on Replica dataset.
- Evaluated on synthetic (Replica) and real-world (TUM-RGBD) benchmarks; demonstrated robustness across varied indoor scenes.

Undergraduate Research Intern *January 2024*
Soft Robotics & Bionics Lab, Dept. of Mechanical Engineering, Seoul National University *Seoul, South Korea*

- Participated in a team project developing a surface electromyography (sEMG) sensor with flexible design elements for enhanced wearability and adaptability.
- Contributed to research on improving sensor stretchability through PDMS (polydimethylsiloxane) and vapour-deposited silver nanoparticles.
- Assisted in applying deep learning models (CNN-GRU, CNN-RNN, ViT) to sEMG signal processing, contributing to efforts to reduce classification error and improve gesture-recognition accuracy.
- Gained proficiency in ML-based signal processing and collaborated with a team of engineers to meet project milestones.

Robotics & Mechatronics Researcher *Aug 2023 – May 2024*
Villanova University *Villanova, PA, USA*

- Collaborated with a Ph.D. candidate on a dissertation titled "Autonomous Localisation and Navigation in GNSS-Denied Environments," advancing SLAM algorithms via LiDAR–camera fusion and visual odometry

for a quad-wheel robot.
- Developed a full ROS-based navigation stack for the robot, implementing path planning and obstacle-avoidance algorithms.
- Applied CNN-based feature extraction and point cloud generation from fused LiDAR–camera data to support autonomous decision-making without GPS.
- Integrated and configured Arduino and Raspberry Pi microcontrollers for robust ROS communication and real-time control.
- Implemented computer vision techniques (edge detection, landmark identification) to build real-time 3D environment maps.
- Implemented a 2D histogram localisation filter and a 1D Kalman filter tracker using probabilistic motion models for improved tracking accuracy.

Professional Experience

Indoor Farm Robotics Intern *June 2023 – Aug 2023*
Area2Farms *Arlington, VA, USA*

- Contributed to the "Silo" automation tool for the local food system supply chain.
- Gained hands-on experience in extruded aluminium construction, pneumatics, industrial robotics, Arduino/Raspberry Pi programming, and agricultural irrigation systems.
- Collaborated with a cross-functional team to prototype automation solutions for indoor vertical farming.

Product Design Intern *June 2022 – Aug 2022*
Ampere LLC (Remote) *Virtual, USA*

- Contributed to 3D product modelling for upcoming consumer technology designs.
- Responsible for analysing structural integrity and physics constraints to ensure product performance and durability.

Distance Education Operator *Sept 2021 – Dec 2021 & Aug 2023 – May 2024*
Villanova University *Villanova, PA, USA*

- Provided technical support for audio/video systems used in online course production.
- Managed live-stream recording and archiving of Distance Education materials for course content reliability.

Collections & Stewardship Technician *Jan 2022 – May 2022*
Villanova University *Villanova, PA, USA*

- Produced digital content for the university Digital Library using scanners, digital cameras, and archiving software.
- Managed records in cataloguing and acquisitions software; supported preservation of rare and special collections materials.

Research Projects

GaussianFeels: Real-Time Multimodal 3D Reconstruction with Tactile-Enhanced Gaussian Splatting (M.S. Thesis) *Dec 2024 – Present*

Objective: Achieve robust in-hand object reconstruction and pose tracking under heavy occlusion by fusing RGB-D vision, DIGIT tactile sensing, and hand proprioception into a shared object-centric 3D Gaussian Splatting representation.

- **Core Representation:** Object-centric Gaussian map updated online from multimodal observations; stored in object coordinates and synchronised to world frame only when rendering or supervision requires it.
- **Population Management:** Contact-aware Gaussian spawning, density boosting at tactile contact regions,

and selective freezing to preserve contact-critical geometry under partial visibility.
- **Pose Tracking:** SLAM-loop tracker combining tactile-aware frame-0 seeding, image-space depth sampling, a Theseus-based SE(3) optimiser, a frozen-map signed-distance residual, and an ICP prior between consecutive depth observations.
- **Manipulation Branch:** Parallel process using Hunyuan3D-2-mini to generate a frame-0 shape prior; progressively replaced by measured camera and tactile geometry across the episode.
- **Evaluation:** FeelSight benchmark family — simulation, real-world rollouts, and occlusion-focused variants; metrics include reconstruction quality, pose-tracking accuracy, and occlusion robustness.

PoP-SLAM: Point Cloud Projection for SLAM (Seongmin Jung, Krishi Attri, Josselin Marchand, Michele Lorenzo Paolicchi) *Sept 2024 – Dec 2024*

- **Innovation:** Projection-first rendering pipeline projecting $\sim$15 000 learned neural points/frame on GPU via vectorised matrix multiplication — no nearest-neighbor search required.
- **Occlusion Handling:** Multi-keyframe depth masking selects only points consistent with measured depth across nearby keyframes — no volume rendering required.
- **Tracking Accuracy:** ATE RMSE 0.75 cm on TUM-RGBD (best among all evaluated methods including Point-SLAM, NICE-SLAM, ESLAM, SplaTAM); 0.38 cm average on Replica.
- **Speed:** $\sim$4 FPS on NVIDIA RTX 4070 (3$\times$ faster than Point-SLAM baseline); $<$3.3% overhead from point pruning.
- **Datasets:** TUM-RGBD (fr1-desk, fr1-room, fr2-xyz, fr3-office) and Replica (synthetic indoor scenes).

Personal & Course Projects

Computer Vision Object Detection Web Application *Summer 2024*

- Built a full-stack web application for real-time image and video object detection using a React frontend and Flask backend.
- Implemented and benchmarked five detection models: Faster R-CNN, Mask R-CNN, RetinaNet, Keypoint R-CNN, and SSDlite using PyTorch and TensorFlow with COCO dataset evaluation. Key findings: Mask R-CNN achieved highest annotation confidence ($>$0.8 scores); SSDlite fastest throughput; Faster R-CNN balanced both.
- Evolved through 8 development versions: from single-model photo processing through modularisation to multi-model video comparison, with an in-development custom "Focus Auto-Population Scan" model.
- Stack: Python (62%), JavaScript (23%), CSS (13%), HTML (1%).

Capstone: Plant Lifting Device for 3D Imaging *Aug 2023 – May 2024*
*Sponsored by a major agrochemical company — **1st Place, Most Innovative Solution***

- Led a multidisciplinary team through the full design-to-prototype lifecycle: actuator selection, motion system (belts, pulleys, drive gears), wiring, waterproofing, and plant platform engineering.
- Addressed sponsor constraints (strict size limits), plant stability risks, and motor-damage risks by developing a stabilising platform and controlled-speed mechanism.
- Applied project management across all design phases; awarded first place for Most Innovative Solution at the university capstone showcase.

Arduino Projects — Mechatronics Coursework *Jan 2022 – May 2022*

- Designed and built three Arduino-based projects integrating pressure, photo, and sound sensors with actuators, motors, and LCD displays.
- Notable project: a puzzle box that randomised solution steps via algorithm and deployed a glitter-spray penalty for incorrect sequences.

Beetle-Bot — Miniature Combat Robot *Aug 2021 – Dec 2021*

- Collaborated in a four-member team to build a miniature combat robot from scratch; achieved 3rd place among sophomore Mechanical Engineering students.

Previous Personal Projects *Summer 2021*

- **Wi-Fi Drone:** Designed and programmed a Wi-Fi-controlled drone with custom stability logic.
- **Swarm Drones:** Developed multi-drone coordination algorithms for communication and synchronisation.
- **Robotic Arm:** Built a servo-actuated robotic arm programmed for pick-and-place operations.
- **3D Printer:** Assembled a 3D printer from scratch including hardware build and slicing software configuration.

AI Mini-Project — Basketball Outcome Prediction *Aug 2020 – Dec 2020*

- Explored ML fundamentals in Python; collaborated in a three-member team to build a predictive model for basketball game outcomes.

ICE Competition — Assistive Device for the Visually Impaired *Aug 2020 – Dec 2020*

*Innovation, Creativity & Entrepreneurship Competition (IIE Idea Challenge) — **3rd Place, College of Engineering***

- Led ideation and business model development in a four-member team, designing a wearable assistive device to aid visually impaired individuals.

Technical Skills

SLAM & 3D Reconstruction

- 3D Gaussian Splatting, Visuo-Tactile SLAM (GaussianFeels), Neural Point Cloud SLAM (PoP-SLAM)
- Point Clouds, Structure from Motion (SfM), RGB-D Reconstruction, Object-Centric Gaussian Mapping
- Contact-aware Gaussian population management, occlusion-aware multimodal supervision, Theseus SE(3) optimisation
- SLAM benchmarking (TUM-RGBD, Replica datasets)

Deep Learning & AI

- PyTorch, TensorFlow, NVIDIA CUDA (custom kernel development)
- Multimodal Learning, Differentiable Rendering
- CNN, RNN, CNN-GRU, CNN-RNN, Vision Transformer (ViT)
- ML for signal processing and predictive analytics
- Reproducible ML workflows, experiment pipeline design, model orchestration
- Ollama (local LLM inference), Genesis (robotics simulation)

Robotics & Middleware

- ROS, OpenCV, Open3D, COCO API
- Sensor Fusion, Real-time Inference, Path Planning, Autonomous Navigation
- Simulation & Sim-to-Real: NVIDIA Omniverse → UR5e + Allegro Hand
- Robot hardware/software integration (perception ↔ actuation APIs)

Hardware & Sensors

- DIGIT Visuo-Tactile Sensor, RGB/RGB-D Cameras, LiDAR (camera fusion)
- VectorNav IMU, Emlid RTK GPS, NVIDIA GPUs
- UR5e Robotic Arm, Allegro Hand, Raspberry Pi / Teensy / Arduino
- 3D Motion Capture (Vicon/OptiTrack)
- 3D Modelling & Printing, Machine Shop Tools

Programming Languages

- Python, C/C++, CUDA, MATLAB, Arduino, LaTeX, HTML, Django, MySQL, VHDL, Maple

Software & Tools

- Simulation & CAD: NVIDIA Omniverse, SOLIDWORKS, Blender, KiCAD

- Dev Tools: Git/GitHub, Jupyter Notebooks, Anaconda, Linux (Ubuntu), PowerShell
- Web: React.js, Flask
- Office & Misc: Microsoft Office, Matplotlib, Sklearn

Languages

- English (Fluent), Hindi (Fluent), Korean (Beginner)

Certifications, Licenses & Awards

NVIDIA Computer Vision Nanodegree *Winter 2024*
Udacity in partnership with NVIDIA

- Modules: Introduction to Computer Vision, Advanced CV & Deep Learning, Object Tracking & Localisation, Cloud Computing, Neural Network Training, C++ Programming, Applied Deep Learning Projects.

Innovative Robot Technologies and their Applications *Aug 2024*
K-MOOC × Seoul National University Credential ID: c65ee02c32ee88873985a799fab29ef8

GRE General Test *Sept 2023*
Credential ID: 82736046 Blockchain ID: 809252

- Verbal Reasoning: 153 (56th percentile)
- Quantitative Reasoning: 166 (80th percentile)
- Analytical Writing: 4.0 (56th percentile)

Capstone Project — 1st Place, Most Innovative Solution *May 2024*
Villanova University Capstone Showcase (Industry-Sponsored)

ICE Competition — 3rd Place *Dec 2020*
Innovation, Creativity & Entrepreneurship Competition, College of Engineering

Beetle-Bot Competition — 3rd Place *Dec 2021*
Villanova University Mechatronics Course

Dean's List *Fall 2020, Spring 2021*
Villanova University

UPenn Robotics Specialisation *Summer 2021*
Coursera in partnership with University of Pennsylvania

- Aerial Robotics, Computational Motion Planning, Mobility, Perception, Estimation and Learning, Capstone.

Relevant Coursework

Seoul National University

- Deep Learning
- Topics in Control and Automation
- Smart Materials and Design
- Precision Metrology and Vision Inspection
- Control Systems 1
- Engineering Research Ethics and Writing Skills
- TA: Mechanical Product Design

Villanova University

- Computer Programming for Mechanical Engineering
- Mechatronics & Microcontrollers and Applications
- Dynamic Systems I & Lab
- Capstone Design I & II

- ME Undergraduate Research I
- Manufacturing Engineering
- Mechanics of Materials / Solid Mechanics & Design I
- Electrical Circuit Fundamentals
- Material Science I
- Thermodynamics
- Engineering Interdisciplinary Project I
- Intro to Computer-Aided Design & Craft (SOLIDWORKS)

Yonsei University (Exchange)

- Heat Transfer, Fluid Mechanics, Mechanical Vibrations, Mechanical Systems Control, Basic Circuit Theory, Probability and Random Variables

Leadership & Activities

Secretary *May 2021 – May 2023*
Villanova CubeSat Club *Villanova University*

- Fostered community engagement in space science with a focus on CubeSats.

Representative & Head of Special Events *Aug 2020 – Aug 2022*
Villanova International Students' Organisation (VISO) *Villanova University*

- Built connections between upperclassmen and incoming international first-year students; organised social events to strengthen community.

Member *May 2021 – Aug 2022*
Society of Asian Scientists and Engineers (SASE) *Villanova University*

- Supported SASE's mission of preparing Asian-heritage scientists and engineers for the global business landscape.

Member *May 2021 – Aug 2022*
American Society of Mechanical Engineers (ASME) *Villanova University*

- Contributed to professional development activities and chapter engineering initiatives.